

Bridging Scales in the Solar Wind: Linking Bulk Plasma States to Turbulence Regimes Using Unsupervised Machine Learning

Papers etc

Cara L. Waters¹ • Christopher H. K. Chen¹ • Mathew J. Owens² ¹Queen Mary University of London • ²University of Reading

Main Result

Bulk and fluctuation regimes in the solar wind are related but not redundant, allowing us to make probabilistic fluctuation forecasts.

Bulk classifications

2 fast types
'typical' compressed

3 slow types
'typical' weak field dense

1 transient

Fluctuation classifications

5 fluctuation types

quiet
structured
Alfvénic
compressive
high amplitude

Motivation

The solar wind is a superposition of plasma populations forming a continuum of states. How to partition such a continuum remains a central unresolved question.

Using speed alone is not sufficient, and introducing measures of heavy ion composition can help to explain the origin of the wind, but does not account for the evolution through the heliosphere.

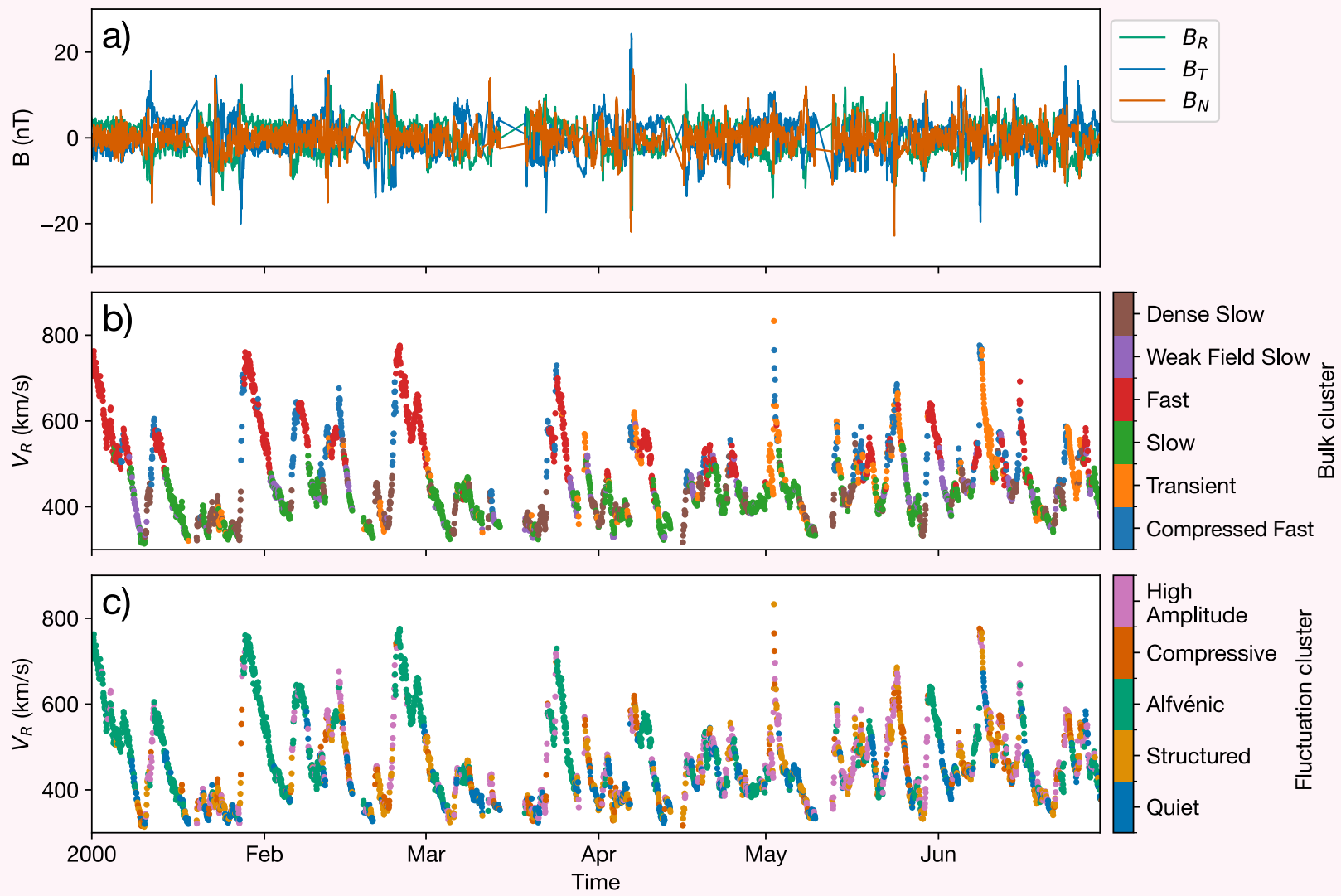
The nature of fluctuations in the wind can vary with type, and we know that such fluctuations can be geoeffective (e.g. Waters+2026). It is therefore important for us to explicitly consider the fluctuation state of the solar wind for space weather purposes.

Data and Methods

We use 14 years of ACE data from 1998 to 2011:
Magnetic field (MAG/MFI), 1 second resolution
Proton data (SWEPAM), 64 second resolution
Heavy ions (SWICS) used for evaluation only with ratio of O⁷⁺/O⁶⁺

Gaussian Mixture Models are used to cluster; one for the bulk properties, and one for the fluctuation properties. This provides a probabilistic measure of both a bulk and fluctuation state.

Fluctuation properties: $|\delta \mathbf{B}_{rms}|$, $|\delta \mathbf{V}_{rms}|$, $\delta N_{rms}/N$, σ_c , σ_r , C_B , $\Delta S_{2.5 \rightarrow 30min}$, $F_{1.5, 10, 30min}$
Bulk properties: V_R , $V_{\perp R}$, $|\mathbf{B}|$, $B_{\perp R}$, T_p , N_p , β , P_{dyn}
All taken as averages over 4 hour rolling windows at hourly cadence



Joint Distribution

We obtain coherency of classifications in time despite providing no explicit time dependence to the clustering. Cluster names assigned by human interpretation of physical properties. Compressed fast wind spread between Alfvénic and high amplitude fluctuations. Transients predominantly contain structured fluctuations. Slow wind quiet in fluctuations, fast wind predominantly Alfvénic. Weak field slow mostly quiet but can be structured or compressive. Dense slow mostly high amplitude. Bulk state is a partial but incomplete predictor of fluctuation regime; both dimensions carry independent information.

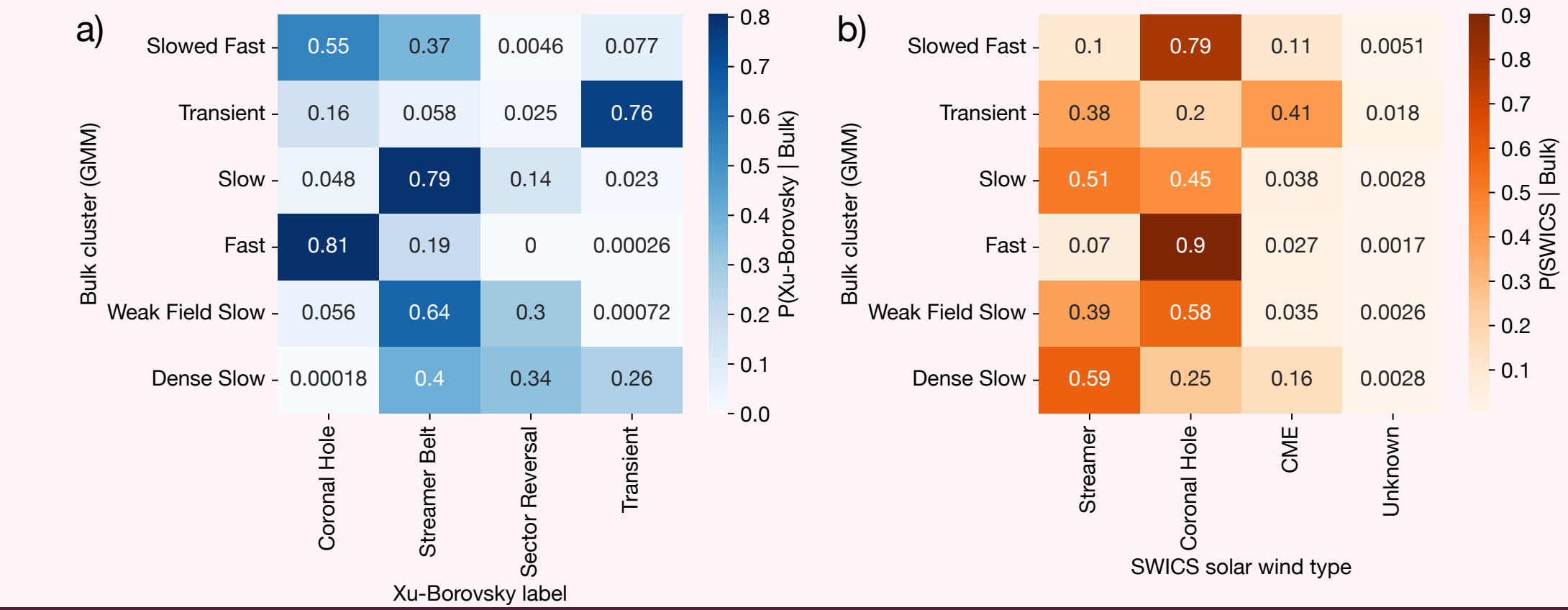
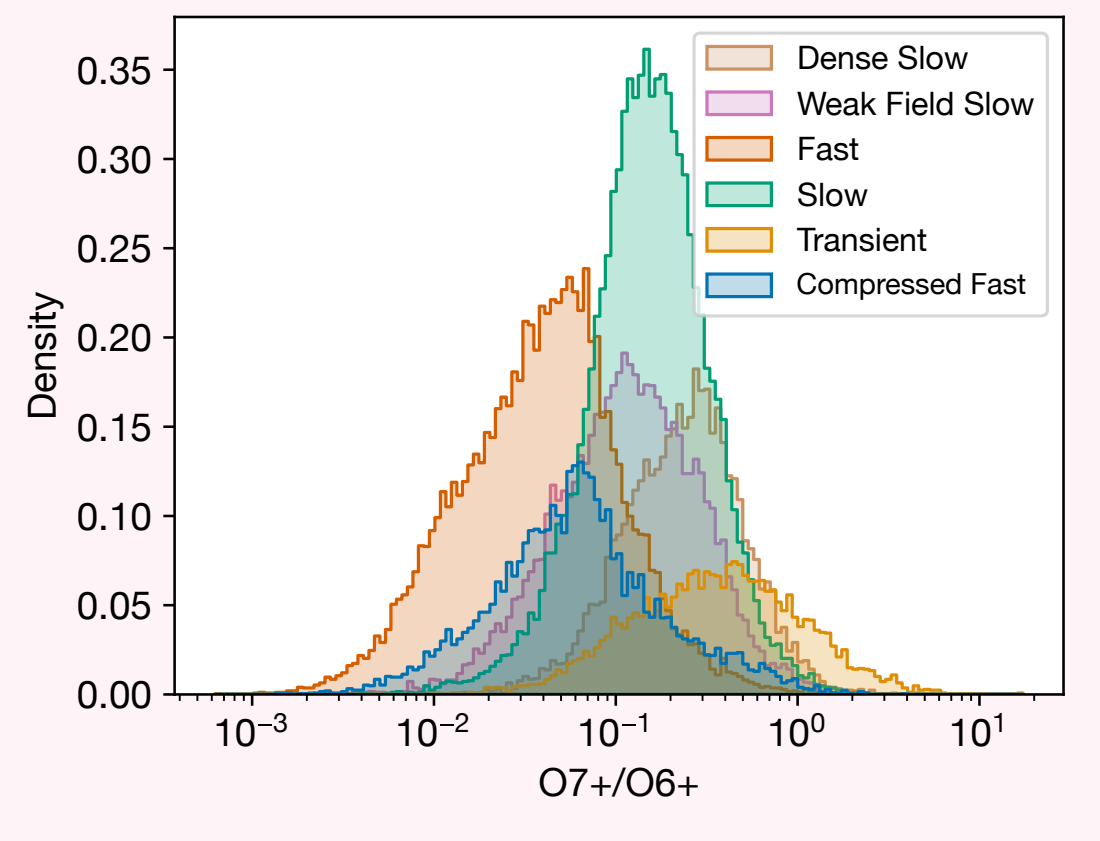


Comparisons

We compare classifications obtained to those of Xu and Borovsky (2014), a method using proton measurements only, and classifications provided with SWICS measurements (Zhao+2009) using speed and O⁷⁺/O⁶⁺ ratio. Fast wind matches well overall. The most ambiguous type is the dense slow, which is spread across streamer/sector reversal and streamer/CME. The correspondence is a reassuring finding for comparison with previous, physics-informed classifications.

Composition

We use the composition, specifically O⁷⁺/O⁶⁺ ratio, to validate against solar origin. Our bulk clusters show some separation consistent with different source regions. When this was compared to other classifications, the overlap was comparable to that of the clusters in the method from Xu and Borovsky (2014). This tells us that the GMM bulk clusters have physically interpretable, source region-consistent identities.



Applications

When models can only provide us with bulk solar wind information, we can use dual classification methods to inform us of the fluctuation properties.

These bulk and fluctuation regimes carry complementary information.

Probabilistic, turbulence-aware classification is a viable and necessary ingredient for space weather relevant fluctuation forecasting.